A different inference space for synthetic gradients

Backup concept for Risk-Calibrated Synthetic Gradients / Compute-Aware Selective Backpropagation

Status: Recorded for later consideration. Not an approved implementation or experiment. The active method, historical results and IMPLEMENTATION_PLAN.md are unchanged.

1. The idea worth preserving

The Jacobi Prior paper by Sourish Das and Shouvik Sardar motivates a change in formulation: choose a quantity with a tractable inference problem, infer it directly, and transform or project the result into the model quantities ultimately needed. Our analogous question is whether a small model of local loss behaviour can replace direct neural prediction of a high-dimensional gradient.

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Prior over local loss behaviour
-> posterior loss surface
-> derived gradient and uncertainty
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2. What the Jacobi construction actually changes

Conventional Bayesian regression places a prior on regression coefficients. The paper instead begins with priors on response-distribution parameters, such as success probabilities. It updates these quantities, transforms their distributions onto a latent scale using a Jacobian, and projects transformed estimates onto the regression design space.

These are still statistical parameters; the change is where uncertainty is represented. The projection is a substantive estimator construction, not just a different prior on the same coefficient model. An induced coefficient distribution does not by itself prove that projecting componentwise modes produces the joint posterior mode of the original regression model.

For this project, the transferable lesson is the choice of inference space and structure. Neither the paper's speedups nor its statistical claims automatically transfer to evolving neural-network gradients. Simply renaming a weight prior or adding a Bayesian risk classifier would miss the stronger reformulation.

3. The proposed object of inference

At a trunk/head split, let h be the representation, y the label and phi the downstream-network state. The downstream loss is f_phi(h,y); the required boundary gradient is its derivative with respect to h.

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g_h = gradient_h f_phi(h, y)
synthetic g_h = gradient_h E[f(h,y) | observations]
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Place a prior over the local function f rather than the weights of a gradient-prediction network. Observe selected loss values and, when affordable, exact derivatives. Infer the local loss surface and derive the gradient afterward. The classifier would still be trained; this idea does not remove its optimization problem.

A tractable construction - and its limits

4. Analytic candidates

Function-space Gaussian model. With a sufficiently smooth fixed kernel and Gaussian observation noise, a Gaussian-process prior yields analytic posterior distributions for function values and derivatives. Loss and derivative observations can be conditioned on jointly. Differentiation of the posterior mean supplies a gradient; differentiation of the covariance supplies uncertainty. Solak et al. provide established precedent for this construction [2].

Bounded local approximation. A fixed low-dimensional representation or a finite quadratic/basis expansion could keep the linear-algebra problem small. Gaussian priors and a linear observation model permit analytic updates. This tractability is conditional on fixed features, covariance/noise choices and observation assumptions; learning those components can reintroduce optimization.

Directional alternative. For a proposed parameter displacement d, model only the loss along the direction: phi(s) = F(theta + s*d). This asks whether a candidate step is worthwhile, without recovering every gradient component. It could serve a gate or step-size decision, but does not by itself generate a full gradient. Probabilistic line searches are a direct precedent [3].

5. Why the formulation might help

Loss-value observations require downstream forward evaluation without downstream backward evaluation. They could provide cheaper information between exact-gradient refreshes. A scalar surrogate also forces its predicted gradients to be derivatives of a common potential; an unrestricted vector predictor need not have that property. Neither advantage is sufficient to establish improved learning.

The relevant comparison is total accuracy versus computation or elapsed time, including target collection, forward probes, exact refreshes, fitting, matrix solves and memory transfers. The current v4 quantile controller is already inexpensive. Any substantial gain must come from useful updates or better allocation, not the mere removal of Bayesian sampling that v4 does not perform.

6. Reasons it could fail

Information loss: a few loss observations cannot identify an arbitrary high-dimensional gradient. Low-dimensional tractability may discard the very directions that matter for learning or harm.

Drift: downstream weights, BatchNorm state, activations and the visited training distribution change. An increasingly confident posterior for an obsolete local function can be misleading.

Cost: repeated forward probes, derivative observations, kernel storage and solves may consume the avoided work. A full Gaussian process growing with all observations is not a cheap default.

Model mismatch: local smoothness, Gaussian noise and the selected feature representation may fail. A posterior credible interval is conditional on these assumptions, not a distribution-free safety guarantee.

Optimization semantics: gradient accuracy or integrability does not establish useful parameter updates after momentum, weight decay and clipping. Head-update and normalization rules must be represented explicitly.

Decision criteria for revisiting the idea

7. Evidence required before adoption

First establish that local downstream-loss behaviour can be predicted from inexpensive information on later batches and changing training states. A frozen-checkpoint fit can diagnose approximation capacity, but cannot validate online calibration or convergence.

Compare the surrogate-derived updates with the current direct predictor and with exact, zero-gradient and skipped-update controls from matched model and optimizer states. Evaluate independent training-only probe loss as well as gradient metrics. A model that is safe only because it makes negligible updates is not a useful replacement.

Charge every observation and refresh to the computation budget. A convincing case requires a better accuracy-computation tradeoff than credible fixed schedules and exact training, allowing a predeclared accuracy loss if savings justify it. Do not infer FLOPs or wall-clock speedup from exact-backprop counts alone.

Use later states and fresh seeds for confirmation after choosing the formulation. Do not reuse examined development outcomes as untouched confirmation data. This note specifies questions for a future proposal, not an authorized experimental matrix or launch instruction.

8. What this idea does not establish

It does not establish a superior predictor, calibrated harmful-update risk, convergence of the production optimizer, final-accuracy guarantees, or novelty by itself. Function-space priors, derivative Gaussian processes and Bayesian directional optimization already exist. A contribution would need to solve the training-specific information, drift and cost problems.

A prior over class probabilities is another possible change of space, but it requires care: uncertainty about the data-generating label probability is not automatically uncertainty about the current network's deterministic output or gradient. For a nonlinear head, uncertainty about its Jacobian also remains. Simple conjugacy for probabilities would not eliminate that computation.

9. Current recommendation

Keep as a backup, not a replacement. Continue the active predictor-utility investigation. Revisit this concept if evidence suggests the direct predictor is difficult to fit or calibrate, while local loss behaviour has exploitable structure. If that structure is absent, the analytic method would be cheaper because it solves an inadequate approximation, not because it solves the original problem efficiently.

Sources and provenance

[1] Das, S. and Sardar, S. *Jacobi Prior: An Alternative Bayesian Method for Supervised Learning.* Uploaded project-root PDF, especially Sections 2-3. This note records an analogy and critical interpretation; it does not endorse all theoretical claims.

[2] Solak et al. *Derivative Observations in Gaussian Process Models of Dynamic Systems.* NeurIPS 15.
Primary paper: function and derivative observations

[3] Mahsereci, M. and Hennig, P. *Probabilistic Line Searches for Stochastic Optimization.* JMLR 18, 2017.
Primary paper: Bayesian inference along an update direction